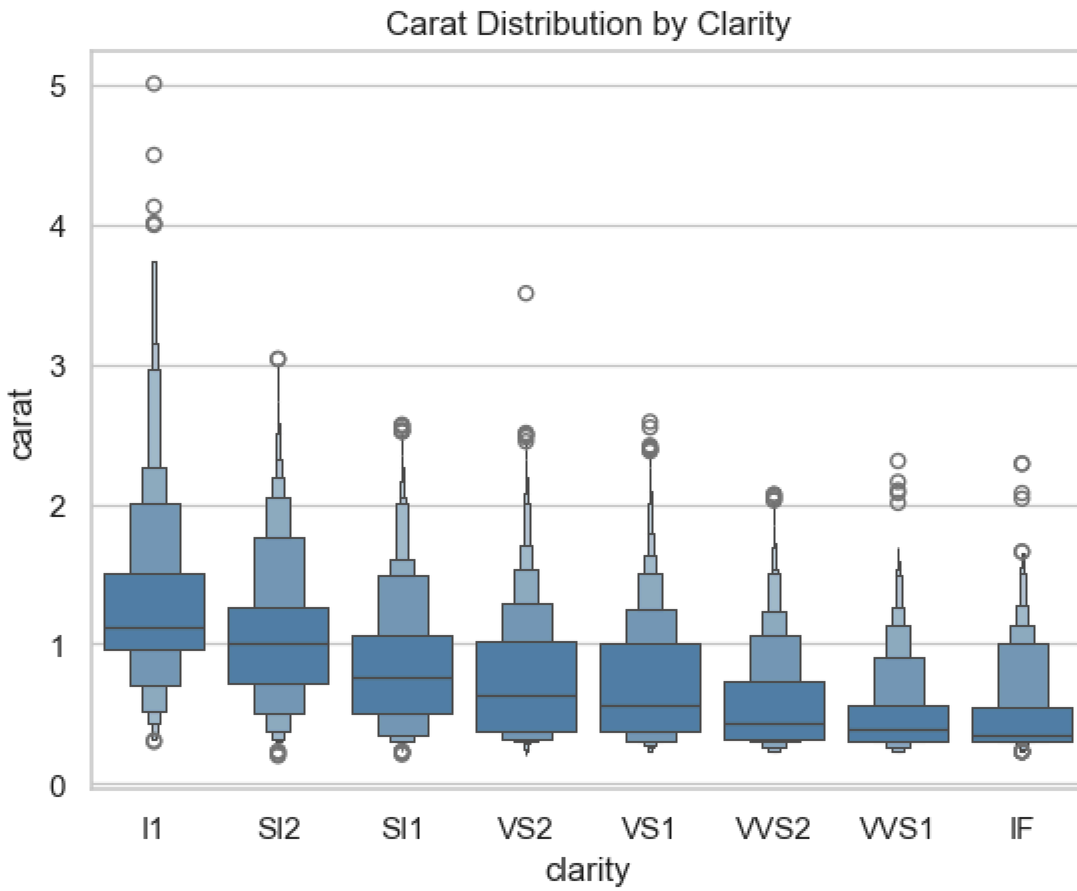
1. pandas
2. numpy
3. matplotlib
4. seaborn
5. plotly
6. scipy
7. scikit-learn
8. streamlit

In [ ]:
```
# pip install pandas numpy matplotlib seaborn plotly scipy scikit-learn streamlit
```

In [9]:
```
import pandas as pd
import numpy as numpy
import matplotlib.pyplot as plt
import seaborn as sns
import plotly as px
```

In [14]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# Theme
sns.set_theme(style="whitegrid")

# Load data
diamonds = sns.load_dataset("diamonds")

# Category order
clarity_ranking = [
    "I1", "SI2", "SI1",
    "VS2", "VS1",
    "VVS2", "VVS1",
    "IF"
]

# Boxen plot
sns.boxenplot(
    x="clarity",
    y="carat",
    data=diamonds,
    order=clarity_ranking,
    color="steelblue"
)

plt.title("Carat Distribution by Clarity")
plt.show()
```



In [10]:
```python
# Read the Excel file
df = pd.read_excel("Python_Practice_Dataset.xlsx")

# Display the first 5 rows
print(df.head())
```

```
   Student_ID    Name Gender  Age Department  Study_Hours  Attendance_%  \
0         101     Ali      M   21         CS           12            90   
1         102    Sara      F   22    Biotech           15            95   
2         103   Ahmed      M   20       Agri            8            80   
3         104  Ayesha      F   23         CS           18            98   
4         105   Usman      M   21   Business           10            85   

   Assignment_Score  Midterm  Final  CGPA  
0                85       78     88  3.45  
1                92       84     91  3.82  
2                70       68     74  2.95  
3                96       90     94  3.95  
4                76       72     79  3.20  
```

In [11]:
```python
data = pd.DataFrame({
    "Name": ["Ali", "Sara"],
    "Age": [25, 30]
})

print(data)
```

```
   Name  Age
0   Ali   25
1  Sara   30
```

In [15]:

```python
import plotly.express as px

# Load the built-in Gapminder dataset
df = px.data.gapminder()

# Create an animated scatter plot
fig = px.scatter(
    df,
    x="gdpPercap",
    y="lifeExp",
    animation_frame="year",
    animation_group="country",
    size="pop",
    color="continent",
    hover_name="country",
    log_x=True,
    size_max=55,
    range_x=[100, 100000],
    range_y=[25, 90]
)

fig.show()
```

In [17]:

```python
import plotly.express as px

# Load the built-in Gapminder dataset
df = px.data.gapminder()

# Create an animated bar chart
fig = px.bar(
    df,
    x="continent",
    y="pop",
    color="continent",
    animation_frame="year",
    animation_group="country",
    range_y=[0, 4000000000]
)

fig.show()
```